

HEALTHCARE DATA ANALYSIS AND DISEASE PREDICTION USING PYTHON

A Data Analysis Report

MILAN DHOUNDIYAL

DATA SCIENCE

TECHGMENT

APRIL BATCH

Acknowledgement

I would like to express my sincere gratitude to my teachers and everyone who supported me during the completion of this project. This project helped me understand how raw healthcare data can be cleaned, analyzed, visualized, and used for machine learning.

Working on this project gave me practical experience with Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. It also helped me understand that the quality of data is very important before drawing conclusions from any analysis.

Abstract

This project focuses on the analysis of a healthcare dataset containing 30,000 patient records. The dataset includes Patient ID, Age, Gender, Disease, Blood Pressure, Sugar Level, and BMI. The aim of the project was to understand the data, identify meaningful patterns, and investigate whether basic patient health information could be used for disease prediction.

The project began with data inspection and cleaning. Missing values were handled, numerical values were converted into suitable formats, and the original Blood Pressure column was separated into Systolic BP and Diastolic BP. A new Blood Pressure Category was then created. Exploratory Data Analysis, statistical analysis, correlation analysis, and risk-factor analysis were performed.

Three machine learning models—Logistic Regression, Decision Tree, and Random Forest—were trained and evaluated. Logistic Regression achieved the highest raw accuracy of 25.57%, followed by Random Forest at 18.95% and Decision Tree at 15.47%. However, the overall classification performance was weak for most disease classes. Age, Sugar Level, and BMI were the most influential features in the Random Forest model. The project therefore presents both the useful patterns and the limitations found in the dataset.

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1. Introduction

Healthcare is an area where data can provide valuable information about patients and diseases. When thousands of patient records are collected together, it becomes difficult to understand the information manually. Data analysis helps organize this information and identify patterns that may not be obvious from individual records.

In this project, Python was used to study a healthcare dataset containing 30,000 patient records. The analysis focused on demographic variables and health measurements such as age, sugar level, BMI, and blood pressure. Along with descriptive analysis and visualization, machine learning models were developed to investigate whether these features could be used to predict disease categories.

An important part of this project was recognizing when the data did not support a strong conclusion. This makes the analysis more realistic and prevents unsupported claims.

2. Problem Statement

Healthcare datasets may contain missing values, incomplete information, inconsistent formats, and categorical variables that cannot be directly used in machine learning. In this project, the raw dataset also contained missing values in several columns and blood pressure stored in a combined format such as systolic/diastolic pressure.

The problem addressed in this project is to clean and prepare the healthcare data, explore its major patterns, study relationships between health variables, and investigate whether patient characteristics can be used to predict disease categories through machine learning.

3. Objectives

- To inspect and understand the structure of the healthcare dataset.
- To identify and handle missing values and inconsistent data types.
- To process Blood Pressure into Systolic BP and Diastolic BP.
- To create a Blood Pressure Category for analysis.
- To study disease distribution and demographic patterns.
- To analyze the distributions of Age, Sugar Level, BMI, and Blood Pressure.
- To perform statistical and correlation analysis.
- To examine possible associations between health parameters and disease groups.
- To train and compare machine learning classification models.
- To identify important predictive features.
- To evaluate the limitations of disease prediction using the available features.

4. Dataset Description

The dataset used in this project contains 30,000 patient records and seven original columns. The data was treated as a messy healthcare dataset so that a complete data analysis workflow could be demonstrated.

Column	Type / Format	Description
Patient ID	Text	Unique patient identifier
Age	Numerical	Age of the patient
Gender	Categorical	Gender category
Disease	Categorical	Disease classification
Blood Pressure	Text	Combined systolic/diastolic value
Sugar Level	Numerical	Recorded sugar level
BMI	Numerical	Body Mass Index

After preprocessing, the dataset included Systolic BP, Diastolic BP, and Blood Pressure Category. The final cleaned dataset contained 30,000 records.

5. Tools and Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Joblib

6. Methodology

The project followed a complete data analysis and machine learning workflow:

Raw Dataset → Data Inspection → Data Cleaning → Feature Transformation → Exploratory Data Analysis → Statistical Analysis → Correlation Analysis → Risk Factor Analysis → Machine Learning → Model Evaluation → Feature Importance → New Patient Prediction

7. Data Cleaning and Preprocessing

The raw dataset contained missing values in Age, Gender, Disease, Blood Pressure, Sugar Level, and BMI. Numerical columns were converted to appropriate numerical formats, while missing numerical values were filled using the median and missing categorical values were filled using the mode.

The Blood Pressure column was split into Systolic BP and Diastolic BP. A new Blood Pressure Category was then created using the project-defined rules. The final distribution was 15,103 Normal, 9,933 High, and 4,964 Low blood pressure records.

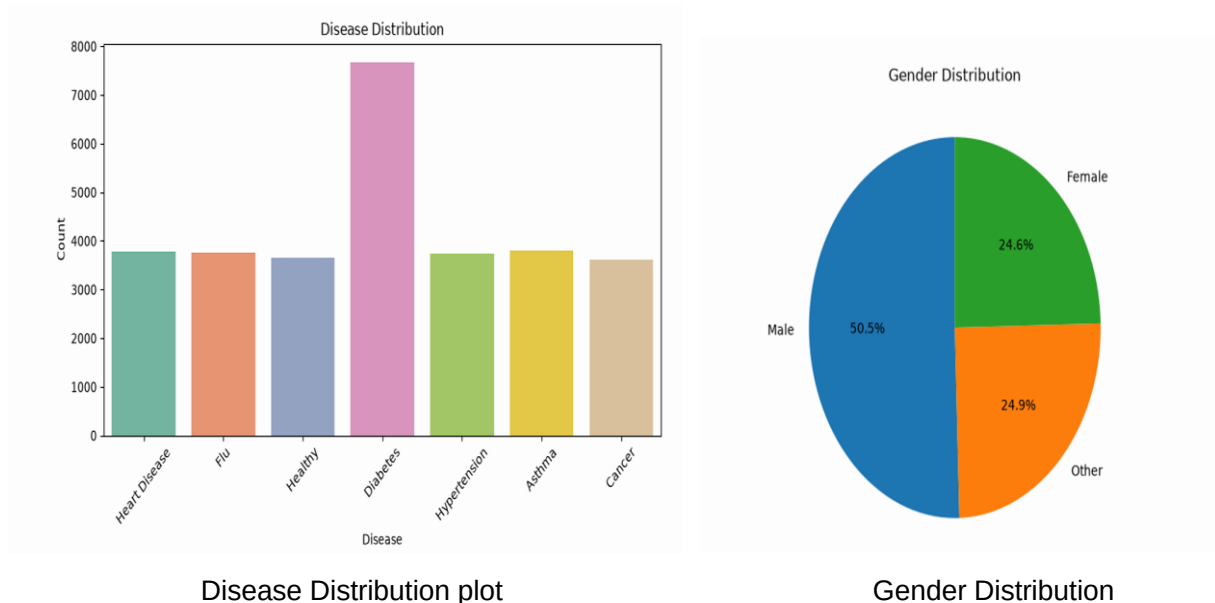
Duplicate checking showed zero duplicate rows. An IQR-based outlier checking step was also included in the workflow, and the final dataset retained 30,000 rows. The cleaned dataset was exported as a separate CSV file for reuse.

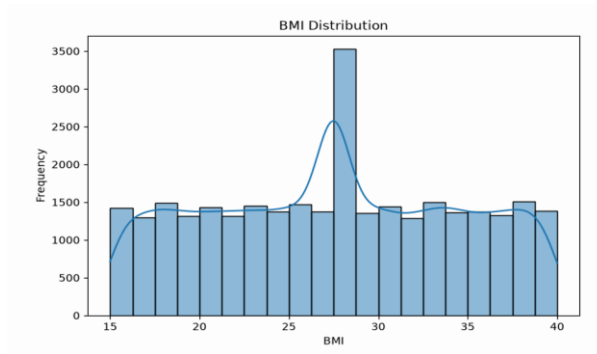
8. Exploratory Data Analysis

Exploratory Data Analysis was used to understand the general structure and patterns in the cleaned dataset. Several visualizations were created, including disease distribution, gender distribution, age distribution, BMI distribution, blood pressure category distribution, sugar level distribution, scatter plots, boxplots, pair plots, and a top-disease chart.

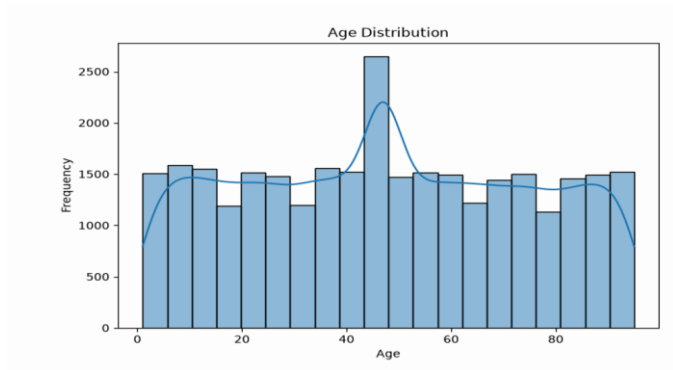
Diabetes was the most common disease category, with 7,671 records, representing approximately 25.6% of the dataset. This imbalance became important during machine learning because a model can appear to perform reasonably well while mainly favouring the largest class.

Gender was relatively balanced, with Male patients representing approximately 50.5% of the dataset. Age ranged from 1 to 95 years, with an average of approximately 47.6 years. The age distribution appeared relatively spread across the full range.

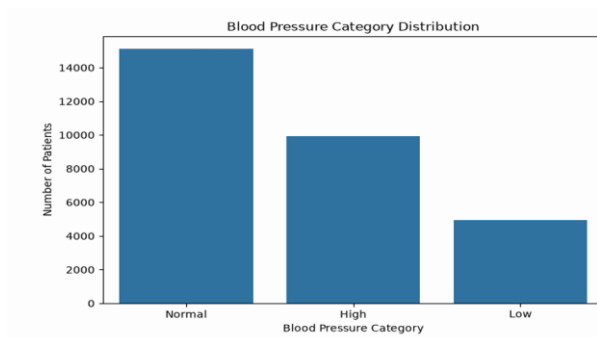




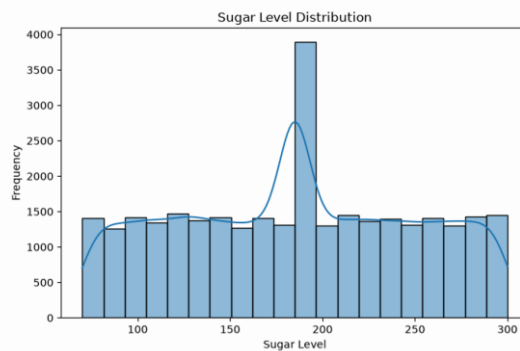
BMI Distribution



Age Distribution



Blood pressure



Sugar level

9. Statistical Analysis

Descriptive statistics were calculated for the main numerical variables. After cleaning, the average age was approximately 47.63 years, the average sugar level was approximately 185.10, the average BMI was approximately 27.50, and the average systolic blood pressure was approximately 128.30.

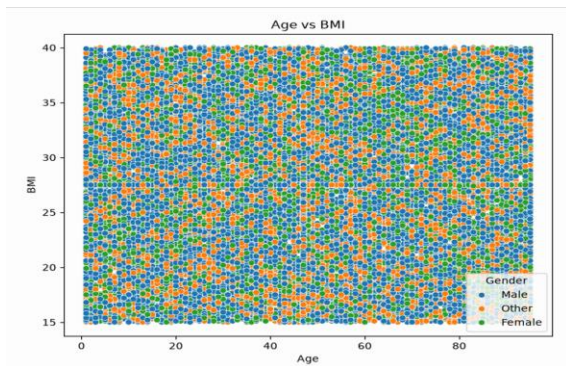
Variable	Mean	Minimum	Maximum
Age	47.63	1	95
Sugar Level	185.10	70	300
BMI	27.50	15	40
Systolic BP	128.30	110	160

These values provide a general overview of this dataset and should not be treated as a description of the general population.

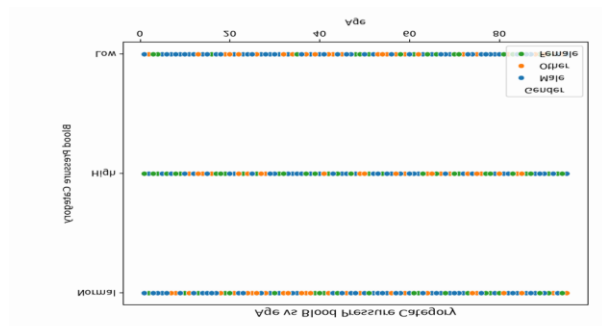
10. Correlation Analysis

Correlation analysis was performed using Age, Sugar Level, BMI, and the numerical form of Blood Pressure Category. The results did not show meaningful relationships between these variables; the correlation values were generally close to zero.

This means that the data does not support statements such as higher age being clearly associated with higher BMI or higher sugar level. The absence of a strong relationship is itself an important result.



Age vs BMI

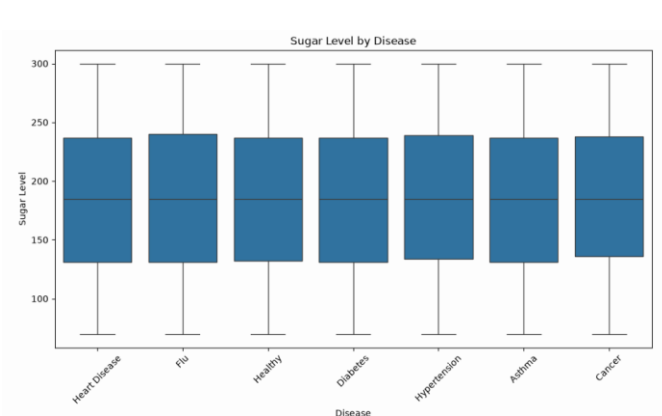
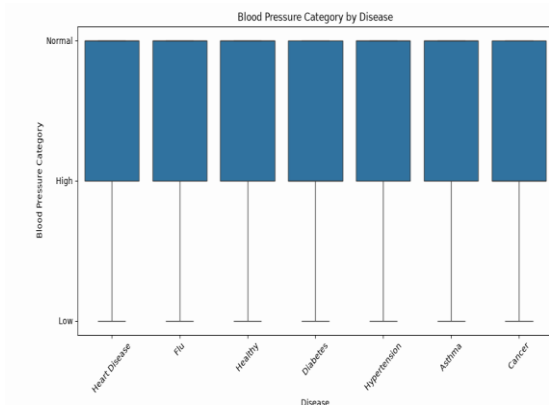


Age vs BP

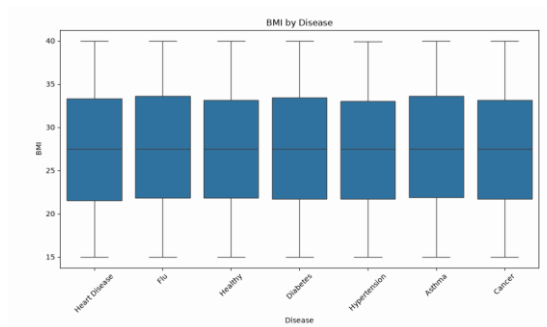
11. Risk Factor Analysis

Risk factor analysis compared BMI, Sugar Level, and Blood Pressure Category across disease groups. The boxplots showed considerable overlap between the disease categories. No disease group clearly separated itself through a consistently higher BMI, sugar level, or blood pressure category.

Therefore, the dataset does not provide strong evidence of a clear disease-specific risk-factor pattern using only these variables. This is a limitation of the available data and not proof that these health factors are medically unimportant.

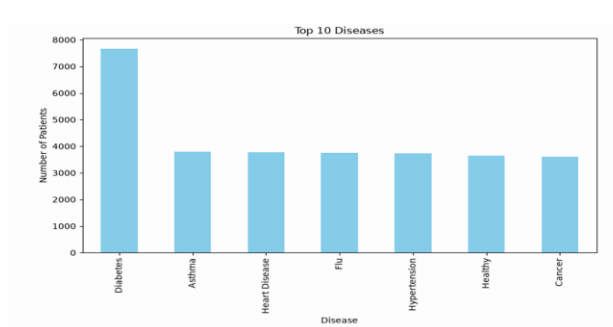


Disease by blood pressure



BMI by disease

Disease by Sugar level



TOP 7 disease

12. Machine Learning

Machine learning was used to investigate whether basic patient information could be used to predict disease categories. The selected features were Age, Gender, BP Category Numeric, Sugar Level, and BMI. The target variable was Disease.

The dataset was divided into 24,000 training records and 6,000 testing records. Gender and Disease were encoded numerically for use by the machine learning algorithms.

13. Model Evaluation

Three classification models were trained and compared: Logistic Regression, Decision Tree, and Random Forest.

Model	Accuracy
Logistic Regression	25.57%
Decision Tree	15.47%
Random Forest	18.95%

Logistic Regression achieved the highest raw accuracy. However, accuracy alone does not tell the complete story because the dataset was imbalanced, with Diabetes representing approximately 25.6% of the records. The Logistic Regression accuracy was therefore close to what could be obtained by heavily favouring the largest class.

The Random Forest classification report showed weak performance for most disease categories. Diabetes was the only class with comparatively better recall at 0.45 and an F1-score of 0.33. The macro-average precision, recall, and F1-score were approximately 0.15, indicating weak performance across the seven-class problem.

The honest conclusion is that the available features did not provide enough strong signal for reliable disease classification.

14. Feature Importance

Feature importance was calculated using the Random Forest model. Age, Sugar Level, and BMI were the most influential features in the model, while Gender and BP Category contributed less to the model's predictions.

These results describe the behaviour of this particular trained model. A feature having lower importance in this model does not mean that it has no medical importance.

15. Patient Disease Prediction

A new patient record was used to demonstrate the prediction process. The model predicted the disease category as Hypertension for the example patient. The Random Forest model was also saved as Healthcare_RF_Model.pkl.

This prediction is strictly an educational demonstration of the machine learning workflow and should not be interpreted as a real medical diagnosis because the overall model performance was weak.

16. Five Meaningful Insights

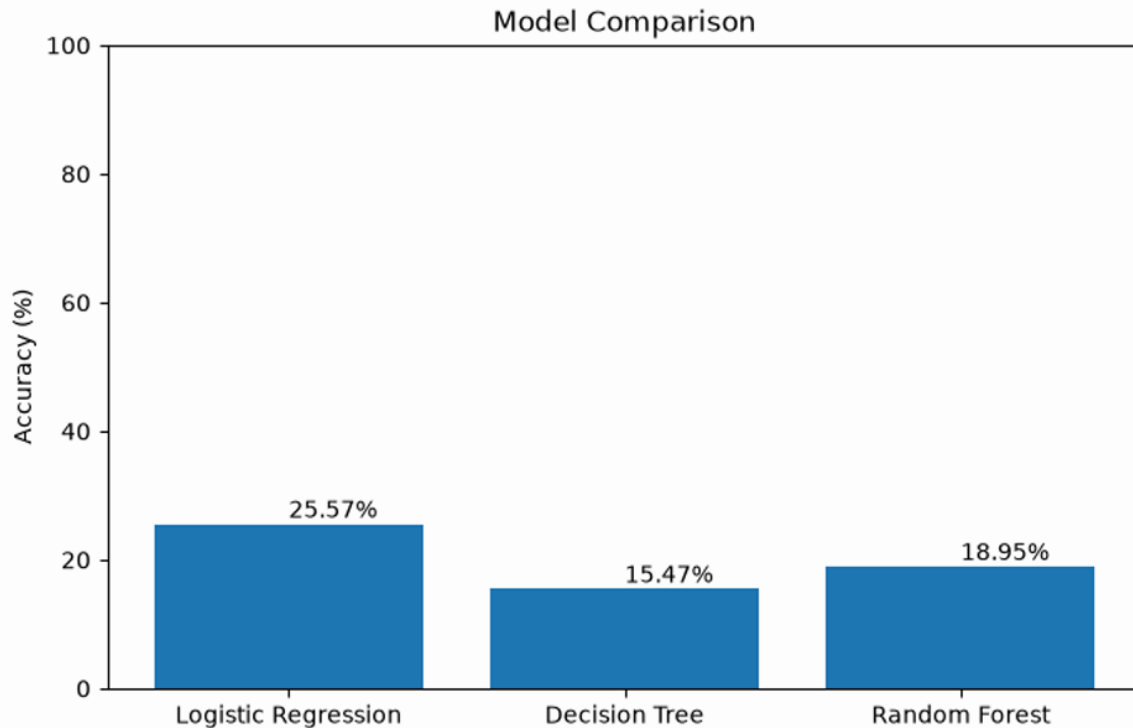
1. Diabetes was the most common disease pattern. Diabetes was recorded 7,671 times, making up about 25.6% of the dataset. It was more common than any other individual disease category. This imbalance later became important when evaluating the machine learning models.

2. The demographic distribution was fairly balanced but not strongly representative of a natural population. Male patients made up about 50.5% of the dataset, and the age range extended from 1 to 95 years with an average of about 47.6 years. The age distribution appeared relatively flat and did not show a strong concentration in one age group.

3. The main health variables did not show meaningful relationships with one another. The correlation analysis showed values close to zero between Age, Sugar Level, BMI, and Blood Pressure Category. Based on this dataset, there is not enough evidence to say that higher age naturally goes together with higher BMI or higher sugar level.

4. No clear disease-specific risk-factor pattern appeared in the visual analysis. The distributions of BMI, Sugar Level, and Blood Pressure Category overlapped considerably across disease groups. The data did not clearly show that one disease group had consistently higher values for these health parameters.

5. The machine learning models struggled to make reliable disease predictions. Logistic Regression had the highest raw accuracy at 25.57%, but this result was close to the proportion of the largest disease class. Overall classification performance was weak, suggesting that the available features did not contain enough useful signal to reliably separate all disease categories. Age, Sugar Level, and BMI were the most influential Random Forest features.



17. Conclusion

This project gave me practical experience in carrying out a complete data analysis and machine learning workflow using healthcare data. I started with a messy dataset and learned how important it is to inspect the data carefully before doing any analysis. Missing values, data types, blood pressure formatting, and categorical variables all needed to be handled before the dataset could be used properly.

The analysis produced five important insights. Diabetes was the most common disease in the dataset, while the gender distribution was fairly balanced and the age distribution was spread across a wide range. The main numerical health variables did not show strong relationships with one another, and the disease groups did not show a clear separation based on BMI, sugar level, or blood pressure category.

The machine learning results were also an important learning point. Although Logistic Regression achieved the highest accuracy of 25.57%, the overall model performance was weak, especially for most disease classes. This suggests that the available features were not sufficient for reliable disease prediction. Age, Sugar Level, and BMI were the most influential features in the Random Forest model, but the model still did not achieve performance strong enough for real clinical use.

Overall, this project helped me understand that a good data analysis project is not only about finding positive results. It is equally important to report when the data does not support a strong relationship or when a model has limitations. In the future, the project could be improved by using more clinically relevant features, better-quality real-world healthcare data, and more advanced modelling and validation techniques.

18. Limitations

- The dataset contains a limited number of health-related variables.
- The disease classes are imbalanced, with Diabetes being the largest class.
- The model performance was weak for most disease categories.
- The available variables may not contain enough clinical information for reliable disease prediction.
- The dataset may not represent the real-world population.
- The results describe statistical patterns in this dataset and cannot be treated as medical conclusions.
- The prediction model is intended for educational purposes and not for clinical diagnosis.

19. Future Scope

The project can be improved by adding more clinically relevant information such as cholesterol, medical history, family history, lifestyle factors, medication history, and laboratory test results. A larger and more representative healthcare dataset could also improve the reliability of the analysis.

Future work could include class-balancing techniques, cross-validation, hyperparameter tuning, advanced machine learning models, and explainable AI methods. The project could also be developed into an interactive healthcare analytics dashboard. Any system intended for real medical use would require careful clinical validation and independent testing before deployment.

20. References

- [Python Documentation](#)
- [Pandas Documentation](#)
- [NumPy Documentation](#)
- [Matplotlib Documentation](#)
- [Seaborn Documentation](#)
- [Scikit-learn Documentation](#)
- Dataset and analysis notebook used for this project